Getting Started with Lesion Tracing: Updated for ATLAS v3.0

Introduction:

Stroke is one of the leading causes of adult disability worldwide, and despite intensive physiotherapy, up to 2/3 of stroke survivors never fully recover. There are still many unknowns about the recovery from stroke, but growing areas such as neuroimaging and big data are helping to inform the field. Many neuroimaging studies are interested in understanding how different properties of lesions (e.g., their size and location in the brain) relate to various aspects of stroke recovery, such as motor recovery after stroke. However, many of the current studies rely on manually traced lesions for their analysis and therefore, the sample size of their analysis might be limited. More recently, techniques have been developed to automatically trace stroke lesions which could allow the potential of running analysis on large datasets. However, there currently are limited ways to test the accuracy of these techniques (i.e. a large set of manually traced lesions reference for comparison). Our goal is to create a large dataset of manually traced lesions in order to have a set guideline for comparing automatically traced lesions techniques and to help provide a training/test dataset for researchers who are trying to improve their automated techniques. We are hoping that this set of manually traced lesions will serve as a useful resource for researchers, providing them with manually traced lesions to assess the accuracy for their own methods. The brains that you will use to trace these lesions are from datasets collected through the ENIGMA Stroke Recovery working group. You can also read prior publications from our lab that have resulted from this manually traced dataset:

- **ATLAS v1.2 publication:** Liew, S.-L., Anglin, J. M., Banks, N. W., Sondag, M., Ito, K. L., Kim, H., ... & Lefebvre, S. (2018). A large, open source dataset of stroke anatomical brain images and manual lesion segmentations. Scientific data, 5, 180011.
- Liew, S.L., Zavaliangos-Petropulu, A., Jahanshad, N., Lang, C.E., Hayward, K.S., Lohse, K.R., Juliano, J.M., Assogna, F., Baugh, L.A., Bhattacharya, A.K. and Bigjahan, B., 2022. The ENIGMA Stroke Recovery Working Group: Big data neuroimaging to study brai
- **ATLAS v2.0 publication:** Liew, S.L., Lo., B.P., Donnelly, M.R., Zavaliangos-Petropulu, A., Jeong, J.N., Barisano, G., … & Yu., Chunshui. (2022). A large, curated, open-source stroke neuroimaging dataset to improve lesion segmentation algorithms. Scientific data, 9, 320. ***– The methods section of this paper contains important information regarding our lesion tracing process which can be used as a reference as you go through this training process.***

Newer publications since the release of the ATLAS v2.0 dataset:

- **IMPORTANT (Our published lesion segmentation protocol):** A standardized protocol for manually segmenting stroke lesions on high-resolution T1-weighted MR images – Frontiers in Neuroimaging, 2023

- Association of Brain Age, Lesion Volume, and Functional Outcome in Patients With Stroke – Neurology, 2023

- Modulation of the Association Between Corticospinal Tract Damage and Outcome After Stroke by White Matter Hyperintensities – Neurology, 2024

- Brain Age Is Longitudinally Associated With Sensorimotor Impairment and Mild Cognitive Impairment in Subacute Stroke – Journal of the American Heart Association, 2025

- Estimating white matter hyperintensities volume in individuals with stroke using T1-weighted images – NeuroImage Reports 2026

Lesion Segmentation Tracing Training:

This training is composed of the following sections:

1. *Software for Lesion Tracing: ITK-SNAP Tutorial*
2. *Stroke Neuroanatomy Training*
3. *Pathology and Lesion Characteristics – Main Things to Remember*
4. *Identifying and distinguishing between Perivascular Spaces (PVS), Lacunae and White Matter Hyperintensities (WMH)*
5. *Common Lesion Tracing Errors to Avoid*
6. *Tracing Checklist Steps*
7. *Manual Segmentation Protocol*
8. *Helpful Links*

1. Software for Lesion Tracing: ITK-SNAP Tutorial:

ITK-SNAP is the software we use for tracing. Complete the following steps to learn how to use it and practice tracing with the different included tools.

- Download ITK-SNAP here: http://www.itksnap.org/pmwiki/pmwiki.php
- Follow sections 1-4 of the ITK-SNAP User Manual: http://www.itksnap.org/docs/fullmanual.php
- Watch these YouTube videos of sample segmentations:
  - https://www.youtube.com/watch?v=ZVmINdWk5R4
  - https://www.youtube.com/watch?v=Iwmdw0XtQKk
- If you feel comfortable with using the software, move onto section 2. If you need more practice, reach out to the tracing team for further guidance.

2. Stroke Neuroanatomy Training:

To learn how to distinguish stroke lesions in the brain, which generally look like atypical structures, it is important to know what a non-lesioned brain looks like. Additionally, some pathology (WMH, PVS) in the brain may look abnormal, but is not a stroke lesion. Pay close attention to these areas.

- **Introduction to neuroanatomy slides**: Accessible from INDI
- **Access instruction videos for specific tracing instructions:**
  1. **Introduction to neuroanatomy, difference between T1/ T2 images, Lobes of the brain:** https://www.youtube.com/watch?v=_3DVfLHWkhI
  2. **Brainstem + Ventricles**: https://www.youtube.com/watch?v=SNCGMd3b_og
  3. **Insula:** https://www.youtube.com/watch?v=QsSeo0vPUas
  4. **Basal ganglia, Perivascular spaces, anatomical planes, slices in ITK Snap:** https://www.youtube.com/watch?v=5nBiJ5FvPxU
  5. **Example lesion tracing walkthrough:** https://www.youtube.com/watch?v=xdnpoVn5caY
     a. Note: this video talks about using interpolation, but we highly recommend drawing the lesion slice-by-slice to avoid missing any lesion anatomy
- Download the first training subject from Training Set 1 and open it in **ITK-SNAP** and practice identifying the following structures (in all three views):
  - Cortical Lobes (temporal, frontal, occipital, parietal)
  - Insula
  - Basal Ganglia (caudate, putamen, pallidum)
  - Brainstem
  - Cerebellum
  - Lateral Ventricle
  - Bonus: try to identify any WMH or PVS!

Use the atlas resources linked at the end of this document for help!

3. Pathology and Lesion Characteristics - Main Things to Remember:

As you look for stroke lesions, keep the following pathological characteristics in mind, and be on the lookout for them:

- Shape (*is it tubular or irregular?*)
- Location (*is this pathology normal to see in this area?*)
- Intensity (*is it very dark (hypointense), or only slightly dark?*)
- Border (*is there a gray halo? (perilesional halo)*)
- Symmetry (*is it occurring in similar areas in both hemispheres?*)
- Direction (*are all shapes 'moving' together, or do they simply 'touch' and then go in different directions/trajectories?)*
- Prevalence (*is it by itself, or are there a lot of similar spots in the brain?)*

Characteristics of…

Stroke lesions:
• Has a perilesional "halo" of medium signal-intensity around the edges, and potentially various signal intensity throughout its shape
• Asymmetrical location and shape
• Impacting nearby gyri/ventricles (appear pushed/shrunken/enlarged/etc.)

Perivascular Spaces (PVS):
• Unlikely to be seen in the brainstem/cerebellum (if it *is*, it will be very clearly following a vascular pattern)
• Generally dark/hypointense
• Very common in the basal ganglia
• Follows the vasculature of the brain
• Stringy, tube-like shape OR round/circular *(depending on which view you're in)*

White Matter Hyperintensities (WMH):
• More homogenous in signal intensity
• Generally symmetrical (occurring in the same area in both hemispheres, ie. "mirrored")
• Very common to see near the horns of the lateral ventricle horns

Characteristics of Lesions in Specific Regions:

Cerebellum

- Fairly rare to see, will likely be obvious

Brainstem/Pons

- Much smaller volume (potentially only a couple slices)
- Likely won't have as noticeable of a perilesional halo (gray border)
- *(Be careful about not tracing vessels coming out of the Sylvian Fissure, which can sometimes look like lesion)*
- *(Lesions here are easier to identify in the Sagittal view)*

Basal Ganglia

- Can follow along a subcortical area (putamen, pallidum)
- Very common to see in this area

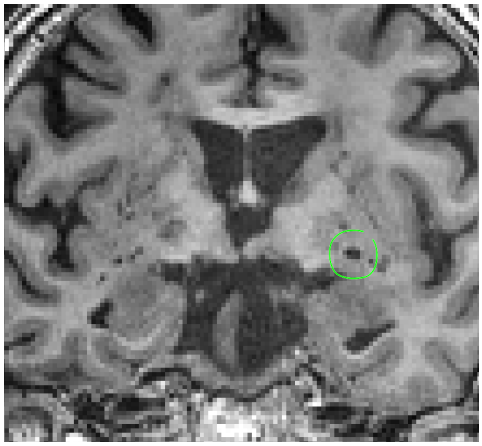
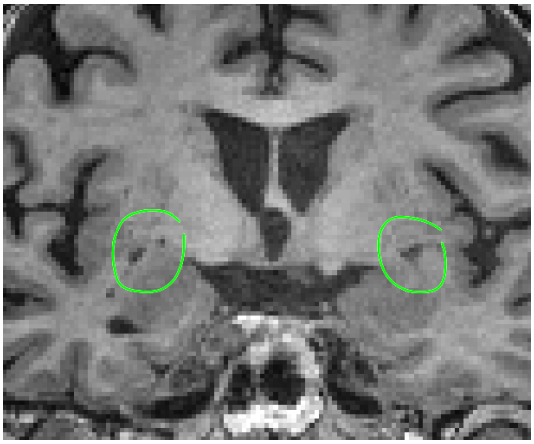
4. Identifying and distinguishing between Perivascular Spaces (PVS), Lacunae and White Matter Hyperintensities (WMH)

Refer to the instructions below on identifying pathology of presumed vascular origin such as PVS and WMH, which are often confused as lesions.

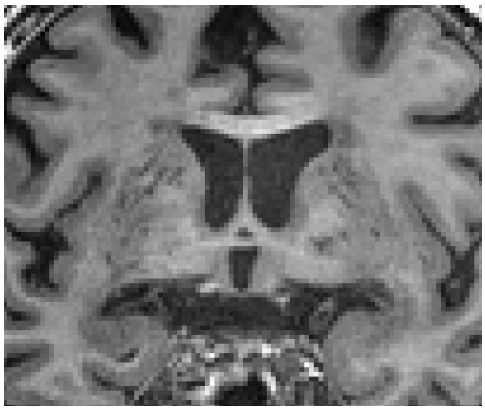
Specific Regions to Look out for PVS in:

PVS from the Sylvian Fissure:

- These are enlarged PVS, not lesions - notice how they come from the Sylvian Fissure

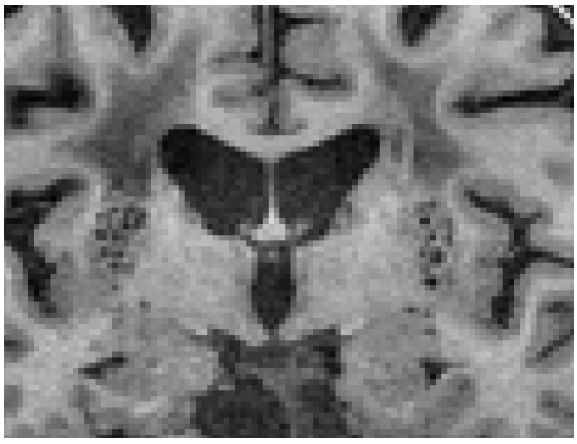
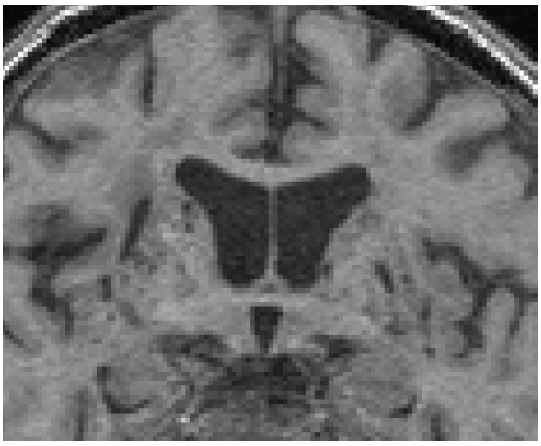



Stringy appearance in the Caudate/Putamen:



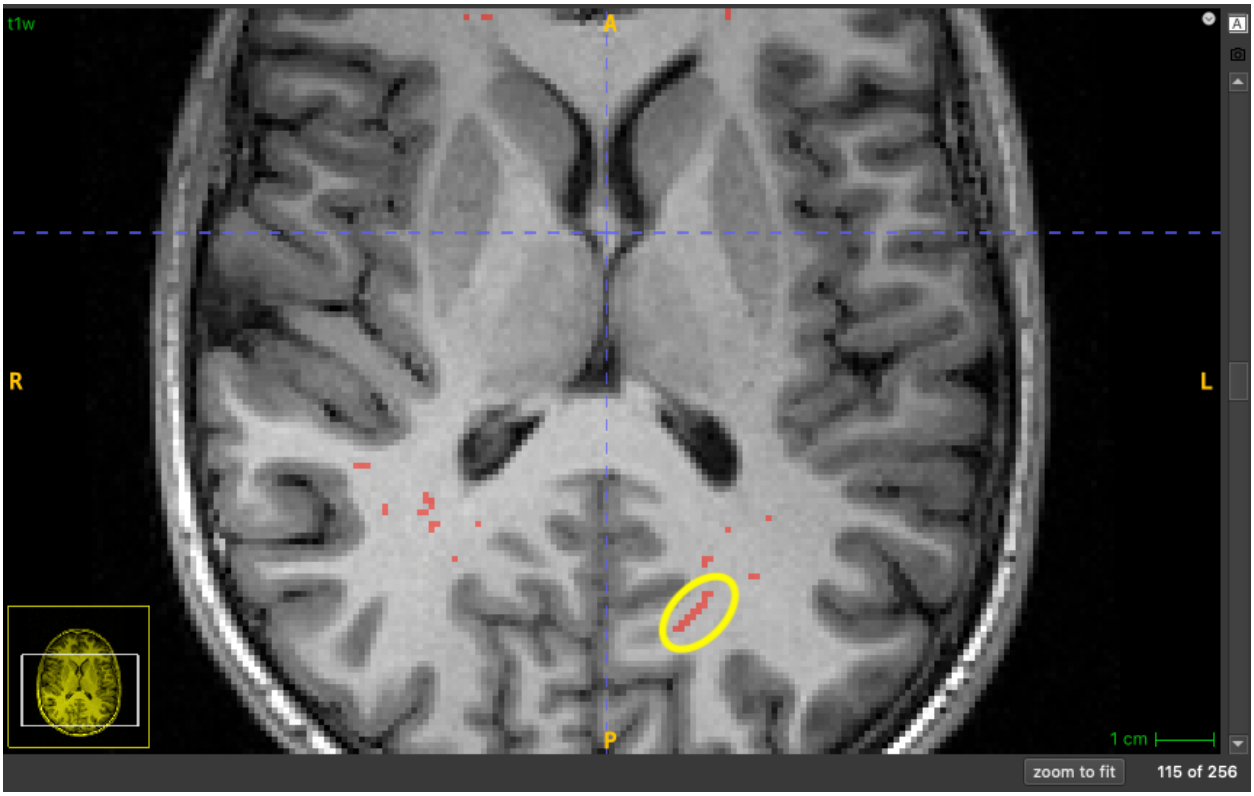
Status Cribrosum:

- Generally associated with high presence of PVS

Distinguishing between WMH and PVS:

An example below where a region of periventricular WMH (which also have presumed vascular origin) might be mistaken as a PVS or lesion.

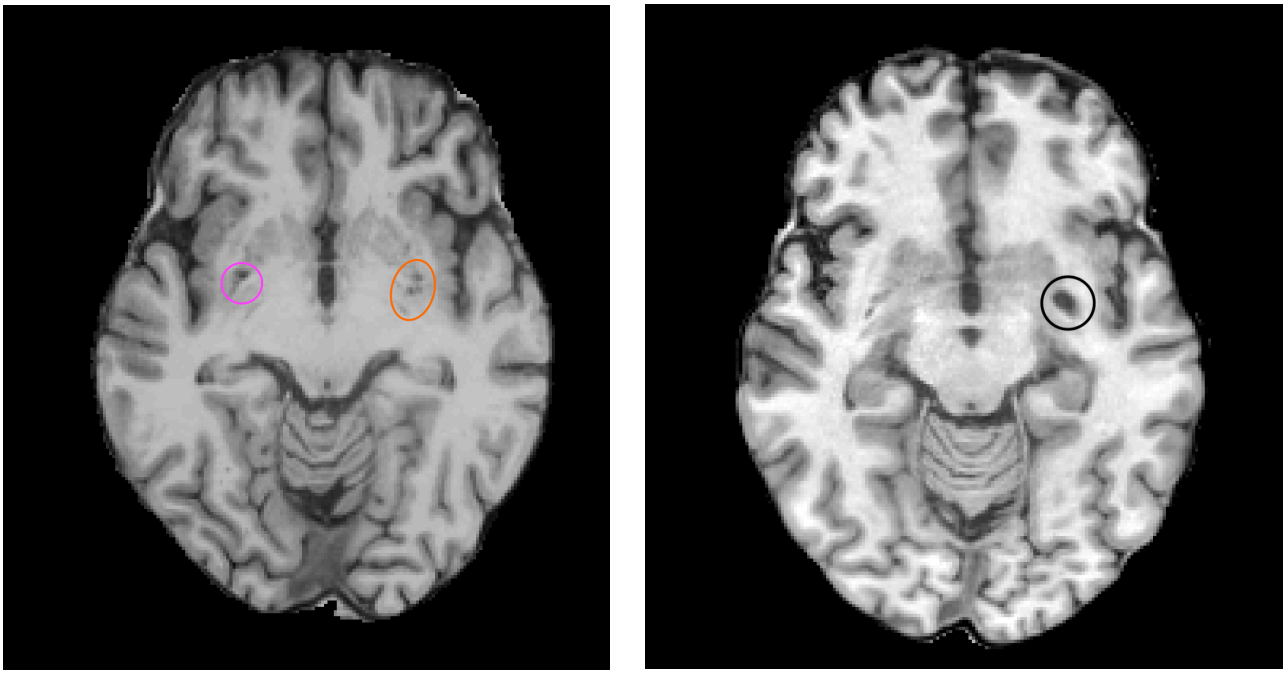


Distinguishing between PVS and lacunae:

In general, a lacune is bigger than PVS (3-15 mm), asymmetric, with irregular shapes (often triangular), and may be located close to a lesion and/or surrounded by hypointense rim (hyperintense on FLAIR). PVS instead tend to be symmetric, smaller than 3 mm, and with regular shapes following the course of blood vessels.

Lacunae: more unstructured, much bigger in size, often extend beyond the basal ganglia
PVS: follow the course of vasculature, much smaller in size, appear commonly in the basal ganglia nuclei

Image A shows multiple smaller PVS (circled in **orange**), and a small lacune (circled in **pink**); Image B shows a larger lacune (circled in **black**).



A **B**

5. Common Lesion Tracing Errors to Avoid:

As you begin identifying and practicing tracing lesions, be mindful of these common tracing errors:

- "*Over-tracing" or "Under-tracing*": Your segmentation border should be right on the edge of where you see the lesion. With the mask at full opacity, you shouldn't see any of the lesion "peeking" out from the mask ("under-tracing"). With the mask at 20-30% opacity, you shouldn't see a brighter red mask border outlining the lesion ("over-tracing").

- *Tracing non-lesioned areas*: If a lesion is near the insula, ventricles, or cortex, the lesion border can be harder to determine. In some cases, it can be helpful to trace what you think is lesion and then go back and "erase" areas that you know are *not* lesion.

- *Asymmetries*: While asymmetries are a good starting indicator that something may be a lesion, head tilt in the scanner can cause the coronal and axial views to have asymmetries that are not lesion. Be mindful of this and be sure to reference all 3 planes.

6. Tracing Checklist Steps:

- Find the T1 nifti file in the finder > right-click “open with” > ITK-SNAP > “Load as Main Image” **DO NOT DOUBLE CLICK FILE
  To Load a Mask:
    a. Open the T1 (same instructions as above)
    b. Open the mask file (click “Load as Segmentation”)
- **Locate lesion** (easier said than done in some cases. If there are multiple lesions, trace all and include in the same mask)
  In the axial (or coronal) view, start posteriorly and scroll through the cerebellum
  Then scroll through and focus on the brainstem
  Then scroll through and focus on the left hemisphere
  Then scroll through and focus on the right hemisphere

1. **Begin tracing lesion:**

    a. May be easiest to start in the center of the lesion (however this is based on preference)
    b. Use one plane/view to start. Choose the plane/view where you see the lesion most clearly. Even if you are tracing in one view, check the other views frequently to see if what you are tracing looks like lesion in those views as well.
    c. Select polygon mode and ‘smooth curve’ option
    d. Outline the lesion by clicking or dragging around the borders of the lesion
    e. Once an outline is made, click complete, then accept if there are no further changes to make; you should now see a filled slice
    f. For medium to large lesions, trace in *either* coronal or axial view, and then go and clean up the borders (if needed) in the *other* view
    g. For large lesions, the ‘interpolate’ tool can be used (but exercise caution in using this, and review the output carefully):
        i. Trace every 2 or 3 slices, repeating until slices have been traced throughout the entirety of the lesion
        ii. Go to Tools > Interpolate Labels (only click ONCE)

2. **When finished tracing, review:**

    a. Turn the mask opacity down to about 20%, and look for any additional lesions
    b. Scroll through in both the coronal and axial views, watching the borders of the mask. Note where there might be any large “jumps” or very straight edges, and fix if needed
    c. Don’t trace *any* of the insula. However, regions of infarct tissue around the insula must be traced.

d. Don't trace *any* of the ventricles
e. Don't trace *any* of the dura mater (white border on the skull)

3. **If you are having trouble determining the lesion border:**

a. Use the crosshairs and click on the border you're unsure about, then look in the other views and scroll through to see if that area is "connected" to anything
b. Look at the general shape and "movement" of the lesion

7. Manual Segmentation Protocol:

Our standardized, manual segmentation protocol has been published and is available here. Before you start tracing lesions independently, the training will happen in 2 phases:

For the first phase, you will first trace 5 practice subjects, receive feedback on those tracings, and update your masks based on the feedback. You will then wait for a week and retrace lesions on those same 5 subjects (with 2 new subjects added). Tracing for both phases are outlined below:

Phase 1:

Follow these steps:

1. Download the set 1 lesions from INDI
2. Trace lesions, following the checklist outlined above
3. When you save your lesion mask, use BIDS-naming, e.g. **sub-r000s000_desc-T1-lesion_mask.nii.gz**
4. Once a trained reviewer has verified and approved your lesions based on the provided gold standard, you must wait for a week before getting started on set 2 lesions.

Phase 2:

1. Download the set 2 lesions from INDI
2. As done for set 1, identify and trace lesions
3. Use the BIDS naming convention to save your segmentation mask
4. Update mask reiteratively based on reviewer feedback until you have been approved based on comparison to the gold standard.

After completing both phases of training and if the checker has fully approved, you are a fully trained lesion tracer and ready to start independently working on data!

Using a Tablet

It can be useful to trace lesions using an electronic pen and tablet. If you have a tablet (Microsoft Surface, iPad, etc.) you may prefer tracing with a stylus/pen. Others prefer using a normal mouse or trackpad. See what works for you.

8. Helpful Links:

ITK SNAP:

Shortcuts:
https://www.itksnap.org/pmwiki/pmwiki.php?n=Documentation.KeyboardShortcuts
S = show or hide the segmentation
Shift+X = show or hide crosshairs
Hold Right-Click + Drag = zoom in and out

Neuroanatomy Atlases:
https://www.med.harvard.edu/aanlib/cases/caseNA/pb9.htm
https://radiopaedia.org/cases/brain-lobes-annotated-mri-1?lang=us
https://jhpmribrainatlas.rcc.uchicago.edu/mri-brain-atlas